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"An Explainable Agentic AI Framework for Security Operations Centers: Enhancing Threat Detection, Incident Response, and Analyst Trust"

A Research Proposal Submitted in Partial Fulfillment of the Requirements for The Degree of Master of Cybersecurity

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Abstract

The increasing number and complexity of cyber assaults is placing growing pressure on Security Operations Centers (SOCs), resulting in alert fatigue, slow response times and a shortage of skilled analysts. Agentic Artificial Intelligence (Agentic AI) offers a promising solution for autonomous threat detection and response, but the "black-box" nature of such systems reduces analyst trust and limits adoption. The research aims to design, implement and evaluate a new Explainable Agentic AI Framework that integrates autonomous AI agents with Explainable AI (XAI) mechanisms. The framework will enhance threat detection, enhance incident response metrics (MTTD/MTTR), and, importantly, build analyst trust via transparent, human-understandable explanations. We will evaluate the framework against traditional and AI-based SOC approaches through a quantitative experimental methodology, using public cybersecurity datasets (CICIDS2017, UNSW-NB15) and a simulated SOC environment.

Keywords

Agentic AI; Explainable AI (XAI); Security Operations Center (SOC); Threat Detection; Incident Response; Analyst Trust; Autonomous Security; Cybersecurity; Multi-Agent Systems; SHAP; MTTD; MTTR

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CHAPTER ONE

INTRODUCTION

1.1 Introduction

The rapid digital transformation of organizations and the growing sophistication of cyber assaults have greatly increased the responsibilities of Security Operations Centers (SOCs). Today's SOCs must continuously monitor for, detect, analyze and respond to an ever-growing volume of security events from sophisticated enterprise environments like cloud platforms, endpoints, networks and Internet of Things (IoT) devices. The growing sophistication and persistence of cyberattacks have exposed the limitations of traditional rule-based security monitoring approaches to deal with the scale and complexity of contemporary threats. To overcome these challenges, Artificial Intelligence (AI) technologies have been more and more employed in SOC operations. Machine learning and deep learning, as well as automated orchestration systems, have enhanced the ability of SOCs to identify anomalies, prioritize alerts and automate incident response activities. Most recently, Agentic Artificial Intelligence (Agentic AI) has emerged as a promising paradigm that allows autonomous and goal-oriented AI agents to perform complex security tasks with minimal human intervention . These intelligent agents are capable of perceiving their environment independently, reasoning about security events, making decisions and executing appropriate response actions.

Agentic AI provides operational advantages, but the transparency and interpretability of the decisions driven by AI is a major challenge. Security analysts often struggle to understand the rationale behind the AI-generated recommendations and the automated actions taken. The lack of explainability can erode trust in artificial intelligence systems and impede their adoption in actual SOC environments. Explainable Artificial Intelligence (XAI) has become a major research area that aims to help explain AI decisions in a way that humans can understand, improving transparency, accountability, and analyst confidence.

In this research, an Explainable Agentic AI Framework for Security Operations Centers is introduced, which combines autonomous AI agents with explainable decision-making mechanisms. The proposed framework aims to provide better threat detection, effective incident response and higher analyst trust by transparent and interpretable security intelligence.

1.2 Background of the Study

Security Operations Centers are the first line of defense in modern organizations against cyber assaults. Traditionally, SOC operations have relied on Security Information and Event Management (SIEM) systems, predefined correlation rules, and manual investigation processes. While effective against known threats, these approaches have serious limitations in dealing with advanced persistent threats, zero-day attacks, insider threats, and sophisticated social engineering campaigns.

new AI-based cybersecurity solutions have emerged that offer new capabilities for automated threat detection and response. Machine learning algorithms can identify anomalous behavior patterns, while Security Orchestration, Automation and Response (SOAR) platforms automate repetitive security tasks. Recent advances in Agentic Artificial Intelligence systems capable of performing complex security operations autonomously have further progressed toward autonomous SOCs.

As opposed to traditional artificial intelligence systems, agentic AI contains autonomous reasoning, planning, adaptation and decision making capabilities. Within the framework of SOC environments, AI agents are capable of observing security events, correlating data from different sources, assessing incidents, suggesting mitigation actions, and executing response procedures autonomously.

But as artificial intelligence systems become more autonomous, concerns arise about transparency, accountability and trustworthiness of standard artificial intelligence systems. Security analysts often are hesitant in trusting AI decisions if they don't know how the AI came to that decision. In recent years, the importance of Explainable AI (XAI) in security environments has been emphasized .

XAI techniques provide explanations that allow analysts to understand why a threat was flagged, how a decision was made, and what factors contributed to the AI's conclusion.

Prior work has explored Agentic AI and Explainable AI independently in SOC environments, but there has been limited work on combining both into a single framework that enables autonomous security operations and analyst trust simultaneously.

1.3 Problem Statement

The increasing complexity and volume of cybersecurity threats have posed significant operational challenges for Security Operations Centers. Traditional SOCs are often bogged down by alert fatigue, slow incident response and over-reliance on human analysts. Agentic AI Provides Opportunities to Automate Security Operations, Enhance Detection and Response. However, most current Agentic artificial intelligence systems are black-box models, providing limited visibility into the decision-making process.

the lack of explainability reduces analyst trust, hampers decision validation and causes problems for regulatory compliance and accountability. Security analysts may be reluctant to rely on the recommendations made by AI or to take autonomous response actions if the reasons behind those recommendations cannot be sufficiently explained or validated.

Thus, an integrated framework is needed that combines the autonomous capabilities of Agentic AI and the transparency of Explainable AI. Such a framework should improve threat detection and incident response performance, and increase analyst trust and acceptance of AI-driven decisions in SOC environments.

1.4 Significance of the Study

This study contributes to both academic research and practical cybersecurity operations.

Academic Significance

1. Draws upon current research on Agentic AI in the field of cybersecurity.
2. Contributes to the growing body of knowledge on Explainable AI in SOC environments.
3. Provides a conceptual framework integrating Agentic AI and XAI.

4. Addresses a research gap regarding analyst trust in autonomous security systems.

Practical Significance

1. Improves accuracy of threat detection in SOC operations.
2. Improves the speed and efficiency of incident response.
3. Reduces analyst workload and alert fatigue.
4. Increases trust and acceptance of AI-assisted security decisions.
5. Supports organizational adoption of autonomous SOC technologies.

1.5 Research Questions

Main Research Question

How can an Explainable Agentic AI Framework improve threat detection, incident response and analyst trust in Security Operations Centers?

Sub-Questions

1. What are the limitations of the existing AI-powered SOC solutions?
2. How can Agentic AI improve threat detection and incident response?
3. How Explainable AI may improve transparency and trust in AI-generated security decisions?
4. What architecture is suitable for integrating Agentic AI and XAI in SOC environments?
5. What is the impact of the proposed framework on SOC performance indicators like detection accuracy, MTTD, and MTTR?
6. What is the effect of explainability on the acceptance of autonomous AI decisions by analysts?

1.6 Objectives of the Study

General Objective

To develop and evaluate an Explainable Agentic AI Framework that enhances threat detection, incident response, and analyst trust in Security Operations Centers.

Specific Objectives

1. To examine the use of Agentic AI in SOC operations.
2. To review Explainable AI techniques relevant to the cybersecurity environments.
3. To develop an integrated Explainable Agentic AI Framework for SOCs.
4. To evaluate the framework's effectiveness in threat detection and incident response.
5. To study the impact of explainability on analyst trust and the decision-making process.
6. To compare the proposed framework with traditional SOC approaches.

1.7 Limitations of the Study

This study is subject to the following limitations:

1. The evaluation will be performed on publicly available cybersecurity datasets and simulated SOC environments.
2. The proposed framework will focus on enterprise SOC operations and may not directly generalize to all industry sectors.
3. Analyst trust measurements may be based on controlled evaluation scenarios instead of long-term operational deployments.
4. The study is mainly aimed at threat detection and incident response processes.
5. Regulatory and legal considerations will be discussed conceptually rather than empirically evaluated.

1.8 Methodology of the Study

The research methodology of this study is quantitative and based on experimental evaluation.

The research process consists of the following stages:

1. Literature review of Agentic AI, Explainable AI, and SOC technologies.
2. Design of the Explainable Agentic AI Framework.
3. Development of autonomous AI agents for threat analysis and response.
4. Integration of Explainable AI mechanisms into agent decision-making workflows.
5. Experimental evaluation using cybersecurity datasets.
6. Performance assessment using metrics such as Accuracy, Precision, Recall, F1-Score, Mean Time to Detect (MTTD), and Mean Time to Respond (MTTR).
7. Analysis of analyst trust and explanation acceptance.

1.9 Thesis Structure

This thesis is organized into six chapters:

Chapter One: Introduction

Presents background, problem statement, objectives, research questions, significance, limitations, methodology and overall structure of the research.

Chapter Two: Literature Review

Review of current literature on Security Operations Centers, Agentic AI, Explainable AI, autonomous SOCs and relevant cybersecurity frameworks.

Chapter Three: Research Methodology

Describes the research design, datasets, tools, framework architecture, experimental procedures and evaluation metrics .

Chapter Four: Framework Design and Implementation

Presents the proposed Explainable Agentic AI Framework Details the architecture, components, and implementation process of the presented framework.

Chapter Five: Results and Discussion

Experimental results, performance assessment, comparative analysis and discussion of the results are reported.

Chapter Six: Conclusion and Future Work

This study summarizes the findings, describes the contributions, presents the limitations, and proposes future research directions.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews the theoretical foundation and previous research on Security Operations Center (SOC), Artificial Intelligence (AI), AI Agents, Cognitive AI Agents, Agentic AI, Autonomous SOCs, Explainable Artificial Intelligence (XAI). In this chapter, we aim to identify current challenges, research progress and common shortcomings in AI-enabled cybersecurity practices. The chapter concludes by stating the research gap and a conceptual framework that guides the proposed investigation.

2.2 Security Operations Centers (SOC)

2.2.1 Definition of Security Operations Centers

A Security Operations Center (SOC) is a centralized unit that continuously monitors, detects, investigates, and responds to cybersecurity events for an organization. SOCs function as the central hub of cybersecurity initiatives by delivering immediate insight into security occurrences and orchestrating incident response efforts.

2.2.2 Functions of a SOC

The primary functions of a SOC include:

- Continuous security monitoring
- Threat detection and analysis
- Incident investigation
- Incident response and containment
- Threat intelligence integration
- Compliance monitoring
- Security reporting

2.2.3 Challenges of Traditional SOCs

Traditional SOCs face several operational challenges:

- Alert fatigue
- High false-positive rates
- Shortage of cybersecurity professionals
- Increasing attack complexity
- Slow response times
- Manual investigation processes

These limitations have encouraged organizations to explore AI-driven solutions for SOC modernization.

2.3 Artificial Intelligence in Cybersecurity

2.3.1 Overview of AI in Cybersecurity

Artificial Intelligence has emerged as a transformative technology in cybersecurity. Artificial intelligence systems can also analyze large volumes of security data, detect anomalous patterns and allow for automated decisions..

2.3.2 Applications of AI in SOCs

AI is commonly applied in:

- Intrusion Detection Systems (IDS)
- Malware classification
- User behavior analytics
- Threat intelligence analysis
- Security automation
- Incident response

2.3.3 Benefits of AI in SOC Operations

AI contributes to:

- Faster threat detection
- Improved detection accuracy
- Reduced analyst workload
- Automated incident prioritization
- Enhanced scalability

2.4 AI Agents

2.4.1 Concept of AI Agents

AI agents are autonomous software entities that can sense their environment, process available data, make decisions, and act to achieve specified goals.

An AI agent typically consists of:

- Perception module
- Reasoning engine
- Decision module
- Action execution component
- Learning mechanism

2.4.2 AI Agents in Cybersecurity

Within cybersecurity environments, AI agents can:

- Monitor security events
- Analyze alerts
- Correlate incidents
- Recommend remediation actions

- Execute automated responses

2.4.3 Benefits of AI Agents in SOCs

- Reduced response time
- Continuous monitoring
- Autonomous decision support
- Scalability across large environments
- Improved operational efficiency

2.5 Cognitive AI Agents

2.5.1 Definition of Cognitive AI Agents

Cognitive AI Agents are an extension of traditional AI Agents, that incorporate capabilities based on human cognitive processes (e.g. perception, reasoning, learning, memory, adaptation and contextual understanding).

2.5.2 Cognitive Capabilities

Cognitive agents typically perform:

- Context-aware reasoning
- Knowledge representation
- Learning from experience
- Adaptive decision-making
- Autonomous planning

2.5.3 Cognitive AI Agents in SOCs

In Security Operations Centers, cognitive agents can:

- Correlate multi-source security data
- Understand attack context

- Detect unknown threats
- Adapt incident response strategies
- Support autonomous cybersecurity operations

2.5.4 Limitations

Despite their advantages, cognitive AI agents face challenges such as:

- Lack of transparency
- Complex decision processes
- Trust issues among analysts
- Explainability limitations

2.6 Agentic AI

2.6.1 Definition of Agentic AI

Agentic AI is a term for intelligent systems that can autonomously pursue objectives, plan adaptively, reason, and take action with limited oversight from humans.

2.6.2 Characteristics of Agentic AI

Key characteristics include:

- Autonomy
- Goal-driven decision making
- Multi-step reasoning
- Adaptability
- Self-directed execution

2.6.3 Agentic AI in SOC Environments

Agentic AI enables:

- Autonomous threat hunting
- Dynamic incident response
- Automated workflow orchestration
- Continuous learning from security events

2.6.4 Benefits and Challenges

Benefits:

- Improved operational efficiency
- Reduced analyst burden
- Faster response times

Challenges:

- Trust and accountability

- Explainability
- Governance and compliance

2.7 Autonomous Security Operations Centers

2.7.1 Evolution Toward Autonomous SOCs

SOC evolution can be classified into:

Level 1: Manual SOC

Level 2: Automated SOC

Level 3: AI-Assisted SOC

Level 4: Semi-Autonomous SOC

Level 5: Fully Autonomous SOC

2.7.2 Autonomous SOC Architecture

Autonomous SOCs combine:

- AI Agents
- SOAR platforms
- SIEM systems
- Threat intelligence platforms
- Automated response engines

2.7.3 Current Limitations

- Dependence on human validation
- Explainability concerns
- Regulatory challenges
- Adversarial attacks against AI systems

2.8 Explainable Artificial Intelligence (XAI)

2.8.1 Definition of XAI

Explainable Artificial Intelligence (XAI) refers to methods that provide understandable explanations for decisions made by artificial intelligence systems.

2.8.2 Importance of XAI in Cybersecurity

XAI helps:

- Improve transparency
- Increase analyst trust
- Support decision validation
- Facilitate compliance requirements

2.8.3 Common XAI Techniques

- SHAP
- TreeSHAP
- LIME
- Rule-based explanations
- Feature importance analysis

2.8.4 Challenges of XAI in SOCs

- Explanation latency
- Explanation instability
- Information overload
- Lack of analyst-centered explanations

2.9 Previous Studies

Study 1 (Sugumar, 2024)

Title:

Next-Generation Security Operations Center (SOC) Robustness: Self-Sufficient Detection and Flexible Incident Management with Cognitive AI Agents

Key Findings:

- Introduced Cognitive AI Agents for SOC operations.
- Improved detection accuracy by 18%.
- Reduced MTTD by 35%.
- Reduced MTTR by 42%.

Limitations:

- Lack of Explainable AI.
- Evaluated in simulated environments.
- Limited human trust assessment.

Study 2 (Javed & Kollwitz, 2024)

Title:

Agentic AI in Security Operations Centers: Proactive Protection of Intelligent Infrastructure against Social Engineering Threats

Key Findings:

- Applied Agentic AI to social engineering defense.
- Emphasized proactive threat prevention.
- Improved autonomous identity verification.

Limitations:

- Conceptual framework only.
- Limited practical validation.
- No explainability mechanism.

Study 3 (Mareedu, 2025)

Title:

Autonomous Security Operations Centers (SOC): AI Agents for Threat Assessment, Mitigation and Coordination

Key Findings:

- Reviewed Autonomous SOC evolution.
- Discussed AI Agents, SOAR, SIEM, and EDR integration.
- Defined the vision of Level 5 Autonomous SOC.

Limitations:

- Explainability remains unresolved.
- Continued dependence on Human-in-the-Loop models.
- Governance concerns.

Study 4 (Lee et al., 2026)

Title:

Practical and Analyst-Acceptable Explainable AI for Supporting Decision-Making in SOCs

Key Findings:

- Developed TreeSHAP-based XAI framework.

- Increased explanation acceptance.
- Reduced explanation latency.

Limitations:

- Focused only on XAI.
- Did not integrate AI Agents.
- Did not address autonomous SOC environments.

2.10 Comparison of Previous Studies

Study	Year	Focus Area	AI Agents	Cognitive AI	Agentic AI	Autonomous SOC	XAI	Analyst Trust
Sugumar	2024	Cognitive SOC	✓	✓	✗	Partial	✗	Limited
Javed & Kollwitz	2024	Agentic AI	✓	✗	✓	Partial	✗	Limited
Mareedu	2025	Autonomous SOC	✓	✗	✓	✓	Limited	Limited
Lee et al.	2026	Explainable AI	✗	✗	✗	✗	✓	✓

Table 1 : Comparison of Previous Studies

2.11 Research Gap

The literature reviewed shows a remarkable advancement of the use of AI Agents, Cognitive AI Agents, Agentic AI, Autonomous SOCs, and Explainable AI in security operations.

However, several gaps remain:

1. Existing Agentic AI frameworks do not incorporate explainability mechanisms.
2. Current XAI solutions do not integrate with autonomous AI agents.
3. Analyst trust remains insufficiently addressed.
4. Analyst confidence is still not being properly addressed.
5. There is limited research that integrates Agentic AI and XAI within a cohesive SOC framework.

Thus, a research gap exists to develop a holistic Explainable Agentic AI Framework that can improve threat detection, incident response, and analyst confidence at the same time.

2.12 Conceptual Framework

The proposed conceptual framework consists of five main layers:

1. Data Collection Layer
    - Network Logs
    - Endpoint Telemetry
    - Threat Intelligence
2. Agentic AI Layer
    - Detection Agent
    - Analysis Agent
    - Response Agent
3. Explainability Layer
    - SHAP / TreeSHAP
    - Decision Explanation Engine

4. Analyst Interaction Layer
    - Alert Review
    - Explanation Validation
    - Trust Assessment
5. Response Layer
    - Automated Response
    - Incident Containment
    - Continuous Learning

The framework combines self-governing decision making with transparent reasoning to improve operational efficiency and analyst confidence in Security Operations Centers.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents the methodology used in developing and evaluating the proposed Explainable Agentic AI Framework for Security Operations Centers (SOCs). It discusses the research methodology, experimental setup, data sources, development tools, framework architecture, explainability aspect, and evaluation metrics used to assess the effectiveness of the proposed system in enhancing threat detection, incident response, and analyst confidence.

3.2 Research Design

This research uses the **Design Science Research (DSR) method combined** with **Quantitative Experimental Research**. Since the main aim is to design, develop, and evaluate an innovative cyber security artifact, the Explainable Agentic AI Framework for SOC operations, the Design Science Research methodology is selected.

The research process consists of six phases:

Phase 1: Problem Identification

- Analyze limitations of traditional SOCs.
- Identify challenges related to:
    - Alert fatigue
    - Slow incident response
    - Limited explainability
    - Low analyst trust

Phase 2: Literature Review

Review existing research on:

- Security Operations Centers
- AI in Cybersecurity
- AI Agents
- Cognitive AI Agents
- Agentic AI
- Autonomous SOCs
- Explainable AI

Phase 3: Framework Design

Design the proposed framework integrating:

- Agentic AI
- Explainable AI
- Human-in-the-loop validation

Phase 4: Framework Implementation

Develop the framework components and instantiate them in a SOC simulation environment.

Phase 5: Experimental Evaluation

Develop cybersecurity data sets and evaluate the effectiveness of the framework.

Phase 6: Result Analysis

Use metrics of trust and performance to measure effectiveness.

Phase	Activity	Months
1	Literature Review & Problem Refinement	1-2
2	Framework Design & Architecture Definition	2-3
3	Implementation of Agentic AI & XAI Layers	3-4
4	Experimental Setup & Data Preparation	4-5
5	Evaluation, Testing, & Data Collection	5-6
6	Analysis, Thesis Writing, & Submission	6-7

Table 2: Proposed Timeline

3.3 Experimental Environment

The proposed framework will be evaluated within a simulated Security Operations Center environment.

Infrastructure Components

Component	**Specification**
Operating System	Ubuntu Server 24.04 LTS
CPU	Intel Xeon / AMD EPYC
RAM	32–64 GB
Storage	SSD
Programming Language	Python 3.12
Database	PostgreSQL
Visualization	Grafana
Development Framework	Django

Security Platforms

The experimental SOC may include:

- SIEM Platform
- SOAR Platform
- Threat Intelligence Platform
- Endpoint Monitoring

Possible open-source tools:

- Elastic SIEM
- Wazuh
- TheHive
- Cortex
- MISP

3.4 Data Sources

The framework will be evaluated using publicly available cybersecurity datasets.

CICIDS2017

Contains:

- DDoS
- Port Scanning
- Brute Force
- Botnet
- Web Attacks

Advantages:

- Widely used benchmark dataset.
- Realistic network traffic.

CICIDS2018

Contains:

- Advanced attack scenarios.
- Modern threat patterns.
- Enterprise network simulations.

UNSW-NB15

Contains:

- Normal traffic
- Fuzzers
- Exploits
- Generic attacks
- Reconnaissance attacks

Security Log Sources

Additional log formats may include:

- Syslog
- Windows Event Logs
- Endpoint Telemetry
- Threat Intelligence Feeds

3.5 Development Tools

Programming Environment

Tool	Purpose
Python	AI Development
Django	Framework Dashboard
PostgreSQL	Data Storage
Docker	Containerization
Git	Version Control

AI Libraries

Library	Purpose
Scikit-Learn	Machine Learning
TensorFlow	Deep Learning
PyTorch	Deep Learning
LangChain	Agent Workflow
CrewAI	Multi-Agent Architecture
OpenAI API / Ollama	LLM Integration

Explainability Libraries

Library	Purpose
SHAP	Explainability
TreeSHAP	Feature Attribution
LIME	Local Explanations

3.6 Proposed Agent Architecture Design

The proposed architecture consists of multiple specialized AI agents.

Agent 1: Threat Detection Agent

Responsibilities:

- Monitor security events.
- Detect anomalies.
- Identify potential threats.

Inputs:

- Network traffic
- Security logs
- Endpoint telemetry

Outputs:

- Threat alerts

Agent 2: Threat Analysis Agent

Responsibilities:

- Correlate alerts.
- Assess severity.
- Classify incidents.

Outputs:

- Risk score
- Incident classification

Agent 3: Response Agent

Responsibilities:

- Recommend mitigation actions.
- Execute automated responses.

Examples:

- Block IP address
- Disable account
- Isolate endpoint

Agent 4: Knowledge Agent

Responsibilities:

- Query threat intelligence sources.
- Retrieve attack context.

Sources:

- MITRE ATT&CK
- Threat Intelligence Feeds

Agent 5: Trust and Explanation Agent

Responsibilities:

- Generate explanations.
- Support analyst decision-making.
- Improve transparency.

Multi-Agent Workflow



3.7 XAI Layer Design

The Explainability Layer is integrated into all AI decisions.

Objectives

- Improve transparency.
- Increase analyst trust.
- Support auditing and compliance.

Components

Feature Importance Module

Uses:

- SHAP
- TreeSHAP

Outputs:

- Most influential attack indicators.

Decision Explanation Module

Provides:

- Why was this alert generated?
- Why was this response selected?
- Which factors influenced the decision?

Analyst Dashboard

Displays:

- Alert
- Risk score
- Recommended response
- AI explanation

Example Explanation

Threat: Brute Force Attack

Confidence Score: 94%

Key Factors:

- Failed Login Attempts: 45%
- Source IP Reputation: 30%
- Unusual Access Pattern: 25%

Recommended Action:

Block Source IP

3.8 Measurement Indicators

The effectiveness of the proposed framework will be evaluated using the following metrics.

Threat Detection Metrics

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Measures overall prediction correctness.

Precision

$$Precision = \frac{TP}{TP + FP}$$

Measures correctness of detected attacks.

Recall

$$Recall = \frac{TP}{TP + FN}$$

Measures detection capability.

F1 Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Measures balance between precision and recall.

3.9 SOC Operational Metrics

Mean Time to Detect (MTTD)

Measures the average time required to identify a threat.

Target

Reduction compared to traditional SOC systems.

Mean Time to Respond (MTTR)

Measures the average time required to respond to incidents.

Target

Reduction through agent automation.

Alert Reduction Rate

Measures reduction in false-positive alerts.

Incident Resolution Rate

Measures percentage of incidents successfully handled.

3.10 Explainability and Trust Metrics

Since your thesis focuses on **Analyst Trust**, these metrics are very important.

Explanation Acceptance Rate (EAR)

Percentage of explanations accepted by analysts.

Explanation Sufficiency Score (ESS)

Measures whether explanations provide enough information for decision-making.

Analyst Trust Score (ATS)

Measured using a Likert-scale questionnaire:

Score	Meaning
1	Strongly Disagree
2	Disagree
3	Neutral
4	Agree
5	Strongly Agree

Example questions:

- I understand why the AI generated this alert.
- I trust the AI recommendation.
- The explanation supports my decision-making.

Explanation Latency

Measures the time required to generate explanations.

Goal:

- Lower than traditional SHAP/LIME implementations.
- Suitable for real-time SOC environments.

3.11 Chapter Summary

In this chapter, we describe the research methodology, including the research design, experimental setup, datasets, development tools, agent architecture, XAI layer configuration, and evaluation metrics. These elements provide the foundation for the implementation and evaluation of the proposed Explainable Agentic AI Framework, which is presented in Chapter Four.

References

- Javed, Y., & Kollwitz, C. (2024). Agentic AI in SOC Operations: Proactively Securing Smart Infrastructure Against Social Engineering.
- Lee, et al. (2026). Practical and Analyst-Acceptable Explainable AI for Supporting Decision-Making in SOCs.
- Mareedu, A. (2025). Autonomous Security Operations Centers (SOC): AI Agents for Threat Triage, Response, and Orchestration.
- Sugumar, R. (2024). Next-Generation Security Operations Center (SOC) Resilience: Autonomous Detection and Adaptive Incident Response Using Cognitive AI Agents.